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Automated classification of atmospheric circulation types for compound flood risk assessment: CMIP6 model analysis utilising a deep learning ensemble

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Automated classification of atmospheric circulation types for
compound flood risk assessment: CMIP6 model analysis utilising
a deep learning ensemblePhilipp Heinrich^{1,*} , Stefan Hagemann¹ and Ralf Weisse² ¹ Regional Land and Atmosphere Modeling, Institute of Coastal Systems—Analysis and Modeling, Helmholtz-Zentrum Hereon, Max-Planck-Straße 1, 21502 Geesthacht, Germany² Coastal Climate and Regional Sea Level Changes, Institute of Coastal Systems—Analysis and Modeling, Helmholtz-Zentrum Hereon, Max-Planck-Straße 1, 21502 Geesthacht, Germany

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E-mail: philipp.heinrich@hereon.de**Keywords:** climate change, machine learning, circulation types, compound events, CMIP6, floodingSupplementary material for this article is available [online](#)

Abstract

The simultaneous occurrence of high river discharges and storm surges represents a substantial hazard for many low-lying coastal areas. Potential future changes in the frequency or intensity of such compound flood events are therefore of utmost importance. To assess such changes, large and consistent ensembles with storm surge and hydrological models are needed, but are hardly available. Often the occurrence of compound flood events is linked to the presence of certain atmospheric circulation types. Future changes in the frequency of such patterns can be directly inferred from available climate simulations. A frequently used classification of atmospheric circulation types are the so-called ‘Großwetterlagen’ by Hess and Brezowsky. In this study, possible future changes in the occurrence of these ‘Großwetterlagen’ were analysed using data from 31 realisations of CMIP6 climate simulations for the emission scenarios SSP1-2.6, SSP3-7.0, and SSP5-8.5. Given the subjective nature of the classification, a deep learning ensemble for the automatic classification was developed and applied. In winter, a higher frequency of the atmospheric pattern Cyclonic Westerly towards 2100 could be inferred as a robust result among all models and scenarios. As this circulation type is potentially associated with compound flooding in some parts of the European coasts, this points towards potentially increasing risks from compound flooding in the future.

1. Introduction

With floods being globally among the most common, expensive, and lethal natural disasters (Jonkman 2005, Hu *et al* 2018), they pose a great threat to the people living in coastal areas (Santiago-Collazo *et al* 2019). Coastal flooding can be caused by various factors, including tides, storm surges, waves, precipitation, and river discharge (Wolf 2009). The simultaneous or successive occurrence of two or more extreme flood drivers, like extreme river discharge and extreme storm surges (Bennett *et al* 2023), can be much more devastating than their individual effects

would be under normal circumstances (Clarke *et al* 2021). They pose an even bigger threat to people living in coastal areas (Santiago-Collazo *et al* 2019), which also contain many big economic centres (Wahl *et al* 2015). These so called ‘compound flood events’ have attracted increased attention from the scientific community in recent years, due to their destructive nature. Sadegh *et al* (2018) concluded that neglecting the interaction of flood drivers could result in a severe underestimation of the flood risk, which can affect critical infrastructures (Khanam *et al* 2021).

Several studies established connections between weather patterns and the occurrence of compound

flood events. Hendry *et al* (2019) found a higher frequency of compound events on the western coast of the UK compared to the eastern coast. They concluded that this phenomenon is due to the fact that storms that generate high skew surges and high river discharge share common features on the western coastline, but not on the opposite shore. Likewise, Heinrich *et al* (2023) demonstrated that the west-facing coasts in northern and central Europe exhibit a higher probability for compound flood events than expected by pure chance. Utilising the Großwetterlagen classification, they identified *Cyclonic Westerly* (WZ) as the common driver for the vast majority of these events. This weather pattern is characterised by strong westerly winds and rainfall. Future changes to such weather patterns can be derived directly from global climate projections, thereby eliminating the need for an additional downscaling step often required for climate change impact studies. The aim of this study is to assess whether there is a systematic change in the frequency of patterns that cause compound flood events.

Two studies, James (2007) and Mittermeier *et al* (2022), have previously focused on the automatic classification of Großwetterlagen, which is based on a subjective classification system. James (2007) proposed an objective classification based on composite plots, where each day was assigned a circulation type based on the composite with the highest correlation after temporal smoothing. In James (2006), they used this method to analyse ensemble runs of the global climate model (GCM) HadGeM1 (Martin *et al* 2006), but found no significant changes to the circulation pattern under climate change. In recent years, convolutional neural networks (CNNs) have made significant advancements in image classification (Liu *et al* 2022b), with potential for weather pattern classification. For example, Liu *et al* (2016) utilised a CNN to detect extreme weather events, such as tropical cyclones, in climate datasets. Mittermeier *et al* (2022) subsequently modified this CNN architecture and used it to classify Großwetterlagen, achieving higher accuracy than the composite method by James (2007). They trained an ensemble of CNNs to analyse future changes in a single-model initial-condition large ensemble (SMILE) of the GCM EC-Earth3 under the Shared Socio-Economic Pathways (SSPs) 3-7.0 scenario (Riahi *et al* 2017). However, their networks strongly underestimated the occurrence rate of WZ. Our goal is to improve on their work by training a CNN with better capabilities for WZ detection through the introduction of a new CNN architecture. This will allow us to assess future changes to this weather pattern caused by climate change.

Our study focuses on the detection of WZ patterns, which are associated with compound flood

events in northern and central Europe, and their future changes due to anthropogenic climate change. First, a novel CNN structure for the classification of Großwetterlagen is presented and evaluated over a reference period. Moreover, we examine the influence of additional parameters on the precision of the network. We then assess the impact of internal variability on the results using the Max Planck Institute Grand Ensemble (MPI-GE). To analyse possible future changes in the frequency of WZ, we utilise 31 GCMs from the Coupled Model Intercomparison Project Phase 6 (CMIP6). We consider three scenarios for each model: SSP1-2.6 (low emission), SSP3-7.0 (medium emission), and SSP5-8.5 (high emission). We use a range of different GCMs to minimise the bias and uncertainty associated with the use of a single model. A comparison of model quality over the reference period is included.

2. Data

In this section, we first provide an introduction to Großwetterlagen. We then present the data used to train our CNN for the classification of WZ. This is followed by an overview of the CMIP6 data used for the analysis of future changes under climate change.

2.1. Großwetterlagen

The classification system of Großwetterlagen was introduced by Hess and Brezowsky (1969) to describe large-scale weather systems over Europe and contains 29 different circulation types (see table 1). It has been utilised in many studies and is likely the most widely used set of circulation patterns over Europe. Moreover, it is important to note that each circulation pattern must persist for a minimum of three days, which has to be taken into account for the classification process. The advantages of the classification system include (i) a very long classification catalogue that begins in 1881 and is continuously updated by the German Weather Service to this day, and (ii) the fact that those circulation types can be associated with consistent and typical local weather conditions (Wapler and James 2015). A potential downside of this classification system is its subjective nature, which introduces an element of human error and makes it challenging to apply to large amounts of simulated data.

2.2. Training data

The training data and methods for our CNN were chosen to be as close as possible to those used in Mittermeier *et al* (2022) to allow a direct comparison of the results. To train our network, we utilised the daily geopotential height of the 500 hPa surface (zg500) and daily mean sea level pressure (mslp) provided by the ERA-20C reanalysis dataset of the

Table 1. A table with acronyms, original, and translated names of the 29 Großwetterlagen according to James (2007). The acronyms are based on the German names.

Index	Acronym	Original German definition	Translated English definition
1	WA	Westlage, antizyklonal	Anticyclonic Westerly
2	WZ	Westlage, zyklonal	Cyclonic Westerly
3	WS	Südliche Westlage	South-Shifted Westerly
4	WW	Winkelförmige Westlage	Maritime Westerly (Block E. Europe)
5	SWA	Südwestlage, antizyklonal	Anticyclonic South-Westerly
6	SWZ	Südwestlage, zyklonal	Cyclonic South-Westerly
7	NWA	Nordwestlage, antizyklonal	Anticyclonic North-Westerly
8	NWZ	Nordwestlage, zyklonal	Cyclonic North-Westerly
9	HM	Hoch Mitteleuropa	High over Central Europe
10	BM	Hochdruckbrücke (Rücken) Mitteleuropa	Zonal Ridge across Central Europe
11	TM	Tief Mitteleuropa	Low (Cut-Off) over Central Europe
12	NA	Nordlage, antizyklonal	Anticyclonic Northerly
13	NZ	Nordlage, zyklonal	Cyclonic Northerly
14	HNA	Hoch Nordmeer-Inland, antizyklonal	Icelandic High, Ridge C. Europe
15	HNZ	Hoch Nordmeer-Inland, zyklonal	Icelandic High, Trough C. Europe
16	HB	Hoch Britische Inseln	High over the British Isles
17	TRM	Trog Mitteleuropa	Trough over Central Europe
18	NEA	Nordostlage, antizyklonal	Anticyclonic North-Easterly
19	NEZ	Nordostlage, zyklonal	Cyclonic North-Easterly
20	HFA	Hoch Fennoskandien, antizyklonal	Scandinavian High, Ridge C. Europe
21	HFZ	Hoch Fennoskandien, zyklonal	Scandinavian High, Trough C. Europe
22	HNFA	Hoch Nordmeer-Fennoskandien, antizykl.	High Scandinavia-Iceland, Ridge C. Europe
23	HNFZ	Hoch Nordmeer-Fennoskandien, zyklonal	High Scandinavia-Iceland, Trough C. Europe
24	SEA	Südostlage, antizyklonal	Anticyclonic South-Easterly
25	SEZ	Südostlage, zyklonal	Cyclonic South-Easterly
26	SA	Südlage, antizyklonal	Anticyclonic Southerly
27	SZ	Südlage, zyklonal	Cyclonic Southerly
28	TB	Tief Britische Inseln	Low over the British Isles
29	TRW	Trog Westeuropa	Trough over Western Europe

ECMWF (2014). The dataset contains an atmospheric reanalysis of the 20th century for 1900–2010. As input data domain we selected the area 30° N to 75° N and –65° W to 45° E (figure 1), as in the subjective classification (Hess and Brezowsky 1969).

The input data were then interpolated to a spatial grid of approximately³ 2.8° × 2.8°. Next, each data variable was standardised using Z-Scores. The labels

used for the classification of the 29 circulation types are based on the Großwetterlagen catalogue by Hess and Brezowsky (1969). Days marked as ‘undefined’ in the catalogue were discarded for training purposes because the CNN cannot learn from them. To ensure maximum comparability, we limited the training of the network to 1900–1980, as done in Mittermeier *et al* (2022).

2.3. CMIP6 GCM data

A set of 31 GCMs from the CMIP6 were used to analyse changes in the frequency of the

³ The spatial resolution of 5° given in Mittermeier *et al* (2022) is an error in the text, according to personal communication with the authors (2023).

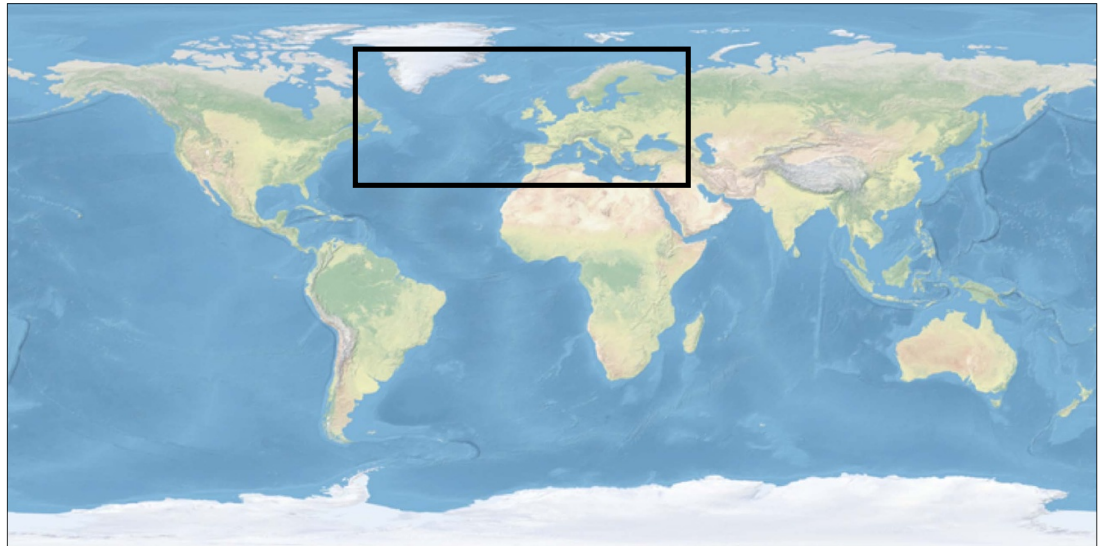


Figure 1. The black rectangle contains the domain used in this study for the analysis of Großwetterlagen. The area covered is 30° N to 75° N and -65° W to 45° E. Map created with cartopy (Met Office 2010) using the Natural Earth shaded relief raster.

Großwetterlage WZ. An overview of those models is given in table 2.

For the selected models, daily data for zg500 and mslp had to be available. Furthermore, it was required that the data for the historical run, as well as the three future scenarios, originate from the same ensemble member. We utilised all models for which data were available via the CMIP6 data search portal of the Earth System Grid Federation at the end of September 2023 and that fulfilled the aforementioned requirements. Additionally, we utilised the AWI-CM-1-1-MR data that were stored on the servers of the German Climate Computing Centre. We chose 1961 to 1990 as the historical reference period because it has been retained as a standard reference period for long-term climate change assessments by the World Meteorological Organisation⁴. For assessing future changes, we utilised the 30 year period from 2071–2100 when available; otherwise the closest period was used that was available for all three future scenarios of a model (see table 2). The datasets were interpolated onto the same grid as the training data to create a consistent input for the CNN. Finally, we standardised the ensemble data using Z-Scores, similar to the training data (see section 2.2).

Furthermore, we utilised daily data from the CMIP6 version of the MPI-GE (Olonscheck *et al* 2023) to examine the contribution of internal variability to the results. MPI-GE is a SMILE of MPI-ESM1.2-LR which consists of 30 realisations for the historical period and five emission scenarios each with CMIP6 forcing. The MPI-GE ensemble members differ in their initial state and the simulated sequence of weather events but share the

same long-term climate statistics. Consequently, differences between the ensemble members are caused only by internal climate variability.

3. Methods: network structure and training process

The Convolutional Neural Meteorology Network (CNMN) employs two input channels for zg500 and mslp data. Our network consists of 4 blocks, each with a single convolution layer with ReLU activation, followed by max pooling, with the last one additionally containing a global average pooling layer (Lin *et al* 2013, Islam *et al* 2021) (see figure 2).

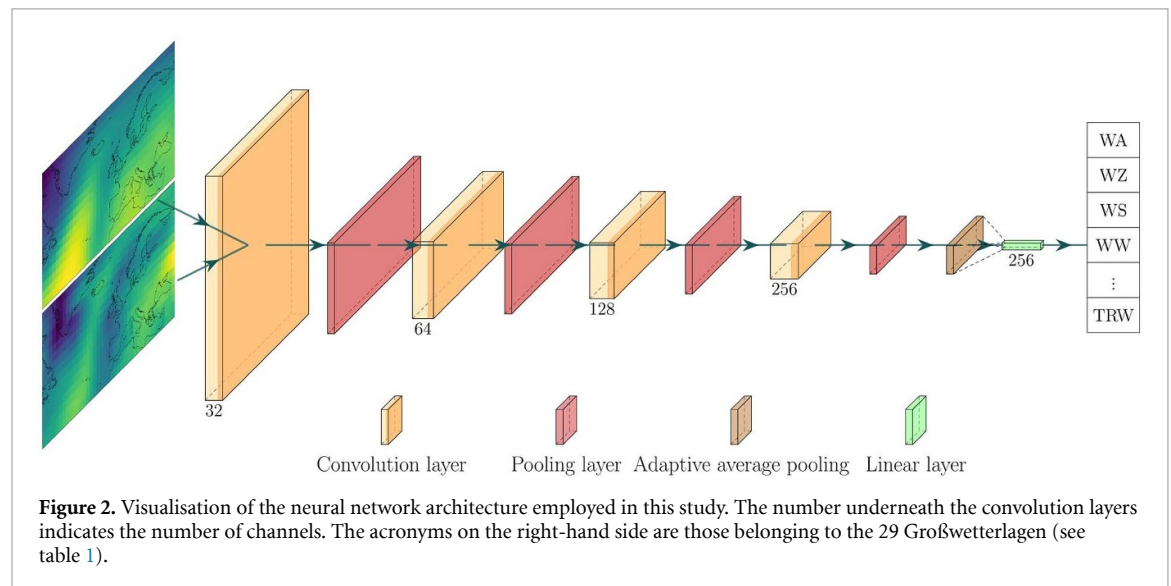
One additional detail is that the convolutional layers use an asymmetrical kernel of size 3×7 instead of a square kernel to account for the larger latitude dimension of the input data. The global average pooling layer collapses the input signal, thereby greatly reducing the size of the subsequent connected layer. This reduced network size helps to mitigate overfitting (Li *et al* 2022). The connected layer then ends in 29 output nodes, one for each Großwetterlage.

The training process used the Adam optimiser (Kingma and Ba 2014) and cross entropy as the loss function. The training was set up for a maximum of 1000 epochs, with an early stop after 100 epochs without improvement and a batch size of 2048. Additionally, Gaussian noise was used for data augmentation to increase the amount of available training data and reduce overfitting. To better assess the performance of CNMN we used k -fold cross-validation, as in Mittermeier *et al* (2022). In this method, the datasets were split into an inner fold containing 70 years and the outer fold containing

⁴ <https://community.wmo.int/en/wmo-climatological-normals>.

Table 2. An overview of the 31 CMIP6 global climate models used in this study. References are based on those given in the data files, or if unavailable, the reference most commonly used by other articles has been included.

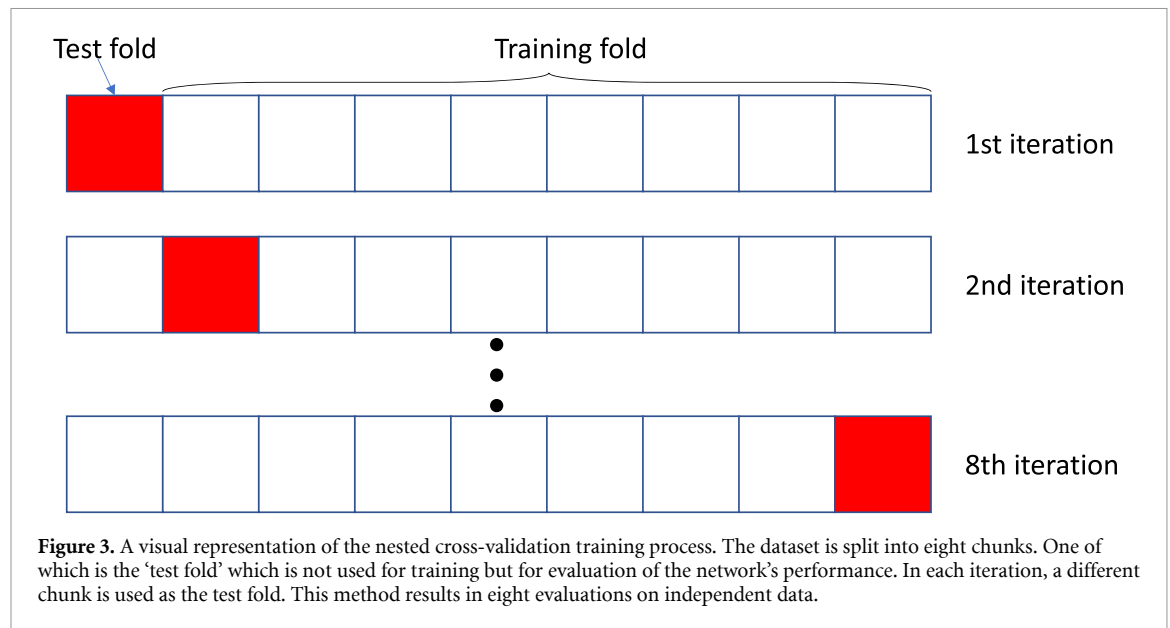
Index	Model name	Variant	Years	Institution ID	Reference
G0	ACCESS-CM2	r1i1p1f1	2071–2100	CSIRO–ARCCSS	Bi <i>et al</i> (2020)
G1	ACCESS-ESM1-5	r10i1p1f1	2071–2100	CSIRO	Ziehn <i>et al</i> (2020)
G2	AWI-CM-1-1-MR	r1i1p1f1	2071–2100	AWI	Tebaldi <i>et al</i> (2021)
G3	BCC-CSM2-MR	r1i1p1f1	2071–2100	BCC	Wu <i>et al</i> (2019)
G4	CanESM5	r10i1p1f1	2071–2100	CCCma	Swart <i>et al</i> (2019)
G5	CESM2	r1i1p1f1	2071–2100	NCAR	Danabasoglu <i>et al</i> (2020)
G6	CESM2-WACCM	r1i1p1f1	2071–2100	NCAR	Gettelman <i>et al</i> (2019)
G7	CMCC-CM2-SR5	r1i1p1f1	2071–2100	CMCC	Cherchi <i>et al</i> (2019)
G8	CMCC-ESM2	r1i1p1f1	2071–2100	CMCC	Cherchi <i>et al</i> (2019)
G9	CNRM-CM6-1	r1i1p1f2	2071–2100	CNRM-CERFACS	Voldoire <i>et al</i> (2019)
G10	CNRM-CM6-1-HR	r1i1p1f2	2071–2100	CNRM-CERFACS	Voldoire <i>et al</i> (2019)
G11	CNRM-ESM2-1	r1i1p1ff	2071–2100	CNRM-CERFACS	Séférian <i>et al</i> (2019)
G12	EC-Earth3	r1i1p1f1	2071–2100	EC-Earth-Consortium	Döscher <i>et al</i> (2022)
G13	EC-Earth3-Veg	r4i1p1f1	2071–2100	EC-Earth-Consortium	Döscher <i>et al</i> (2022)
G14	EC-Earth3-Veg-LR	r1i1p1f1	2071–2100	EC-Earth-Consortium	Döscher <i>et al</i> (2022)
G15	FGOALS-g3	r1i1p1f1	2071–2100	CAS	Li <i>et al</i> (2020)
G16	IITM-ESM	r1i1p1f1	2069–2098	CCCR-IITM	Swapna <i>et al</i> (2015)
G17	INM-CM4-8	r1i1p1f1	2071–2100	INM	Volodin <i>et al</i> (2018)
G18	INM-CM5-0	r1i1p1f1	2071–2100	INM	Volodin <i>et al</i> (2018)
G19	IPSL-CM6A-LR	r1i1p1f1	2071–2100	IPSL	Boucher <i>et al</i> (2020)
G20	KACE-1-0-G	r1i1p1f1	2071–2100	NIMS-KMA	Lee <i>et al</i> (2020a)
G21	MIROC-ES2H	r1i1p4f2	2071–2100	MIROC	Kawamiya <i>et al</i> (2020)
G22	MIROC-ES2L	r1i1p1f2	2071–2100	MIROC	Hajima <i>et al</i> (2020)
G23	MIROC6	r3i1p1f1	2071–2100	MIROC	Tatebe <i>et al</i> (2019)
G24	MPI-ESM1-2-HR	r1i1p1f1	2071–2100	MPI-M	Müller <i>et al</i> (2018)
G25	MPI-ESM1-2-LR	r4i1p1f1	2071–2100	MPI-M	Mauritsen <i>et al</i> (2019)
G26	MRI-ESM2-0	r4i1p1f1	2071–2100	MRI	Yukimoto <i>et al</i> (2019)
G27	NorESM2-LM	r1i1p1f1	2071–2100	NCC	Seland <i>et al</i> (2020)
G28	NorESM2-MM	r1i1p1f1	2071–2100	NCC	Seland <i>et al</i> (2020)
G29	TaiESM1	r1i1p1f1	2071–2100	AS-RCEC	Lee <i>et al</i> (2020b)
G30	UKESM1-0-LL	r8i1p1f2	2071–2100	MOHC	Tang <i>et al</i> (2019)



10 years. The inner fold was utilised for training and tuning the learning rate of our network, while the evaluation of the network was performed on the outer fold. This procedure was then repeated several times for different splits of the data (see figure 3) to obtain

reliable metrics for CNMN. To ensure consistency with the 3 d rule, a separate algorithm was employed (see section S1 for details).

To evaluate the performance of our network, we assessed both the macro *F1* score and overall



accuracy. We trained an ensemble of 50 networks in order to minimise the variability of the network performance caused by the random initialisation of the network weights at the beginning of the training process.

4. Results

4.1. Network evaluation

The performance of our network ensemble is evaluated by comparing its predictions to the actual daily Großwetterlagen in the catalogue, which is based on observational data. This evaluation is conducted on the eight outer test folds, which are independent of the training process. To increase the robustness of our evaluation, we use five different networks for each outer fold to account for the variance caused by random initialisation. Table 3 presents the performance comparison of CNMN with those of Mittermeier *et al* (2022) and James (2007).

Our results show an increase in the macro $F1$ score by almost 2.2% and an accuracy increase of over 7% compared to the previous best results. Our ensemble of CNMNs outperforms the ensemble of Mittermeier *et al* (2022) in 22 out of the 29 Großwetterlagen and the composite method of James (2007) in 20 out of 29 Großwetterlagen. When compared to both studies, CNMN offers the best performance metrics for more than half of the weather constellations. In particular WZ, the primary focus of this study, exhibits significant improvements compared to the classifications in previous studies, with increases of +16.04% and +10.4%, respectively.

4.2. MPI-GE

First, we compare the number of days classified as WZ in the historical period for the MPI-GE ensemble members and those found in the ERA-20C dataset

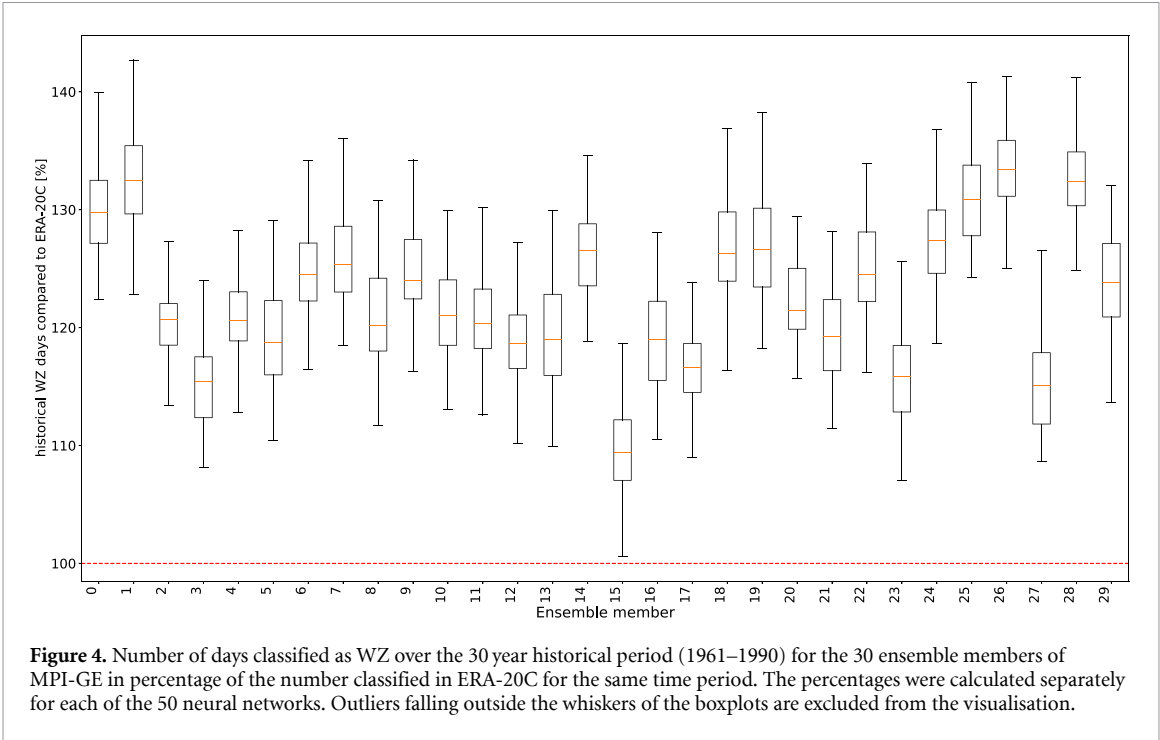
for 1961–1990 (figure 4). The analysis indicates that all MPI-GE ensemble members differ in the number of WZ days by approximately +23% on average over the 30 year period. The uncertainty introduced by the random initialisation of the 50 networks accounts for only 16% of the total uncertainty, while the internal climate variability of 30 MPI-GE ensemble members has a significantly greater impact. These calculations are based on the differences in variability for one network and all MPI-GE ensemble members, as well as the variability observed for 50 networks and the same MPI-GE ensemble member.

For the analysis of changes in the future (2071–2100), we divided the year into two seasons: summer (16 April–15 October) and winter (16 October–15 April). Figure 5 reveals that both seasons show different trends under climate change. During winter, there is a growing number of ensemble members projecting an increase in WZ days for stronger emission scenarios. For SSP1-2.6, 20 out of 30 members project an increase in WZ days, which rises to 28 members for SSP5-8.5. On average, the number of WZ days increases by 2.01 d (SSP1-2.6), to 4.07 d (SSP3-7.0), and lastly, up to 5.70 d (SSP5-8.5) per winter season.

The summer half-year exhibits no discernable trend, with a slight majority of the members favouring a minor decrease in WZ days. For SSP1-2.6, half of the members project an increase in days with WZ in the summer season, while the other half show a decrease, resulting in an average change of +0.2 d throughout the ensemble. For both SSP3-7.0 and SSP5-8.5, the ensemble members are similarly divided, with roughly half the members supporting each type of change. This results in an average close to zero, with a difference of −0.5 d under the strongest climate change scenario, SSP5-8.5. There is notable variability among the members, even within the same model, for both future changes and the historical time

Table 3. A table with the results of our network ensemble, as well as those of Mittermeier *et al* (2022) and James (2007). The F1 score for James (2007) is based on their results (table 2 in the corresponding publication) for classification of ERA40 data from September 1957 to August 2002. The calculations were performed without days that were labelled ‘undefined’.

Index	Acronym	CNMN F1 score (%)	Mittermeier F1 score (%)	James F1 score (%)
1	WA	39.58	44.60	40.32
2	WZ	63.12	47.08	52.69
3	WS	46.72	45.39	35.10
4	WW	40.82	37.70	30.11
5	SWA	29.78	35.36	36.90
6	SWZ	33.50	30.86	39.78
7	NWA	37.42	38.88	33.76
8	NWZ	40.33	37.07	43.57
9	HM	59.07	51.24	43.24
10	BM	51.73	47.29	38.05
11	TM	39.12	37.23	37.08
12	NA	17.00	24.85	15.96
13	NZ	46.22	44.32	41.55
14	HNA	54.63	45.57	45.82
15	HNZ	28.98	27.11	37.02
16	HB	52.72	50.99	44.94
17	TRM	29.96	27.86	39.58
18	NEA	43.19	41.44	30.02
19	NEZ	36.31	33.12	27.27
20	HFA	51.50	45.32	41.13
21	HFZ	24.13	24.81	32.89
22	HNFA	33.44	33.35	43.48
23	HNFZ	36.13	34.02	33.06
24	SEA	34.72	38.09	27.24
25	SEZ	36.20	37.93	31.10
26	SA	42.81	39.84	34.04
27	SZ	43.49	38.19	26.67
28	TB	47.30	42.11	38.00
29	TRW	33.96	29.34	37.85
Mean macro F1 score (%)		40.48	38.30	36.49
Accuracy (%)		48.33	41.10	39.51



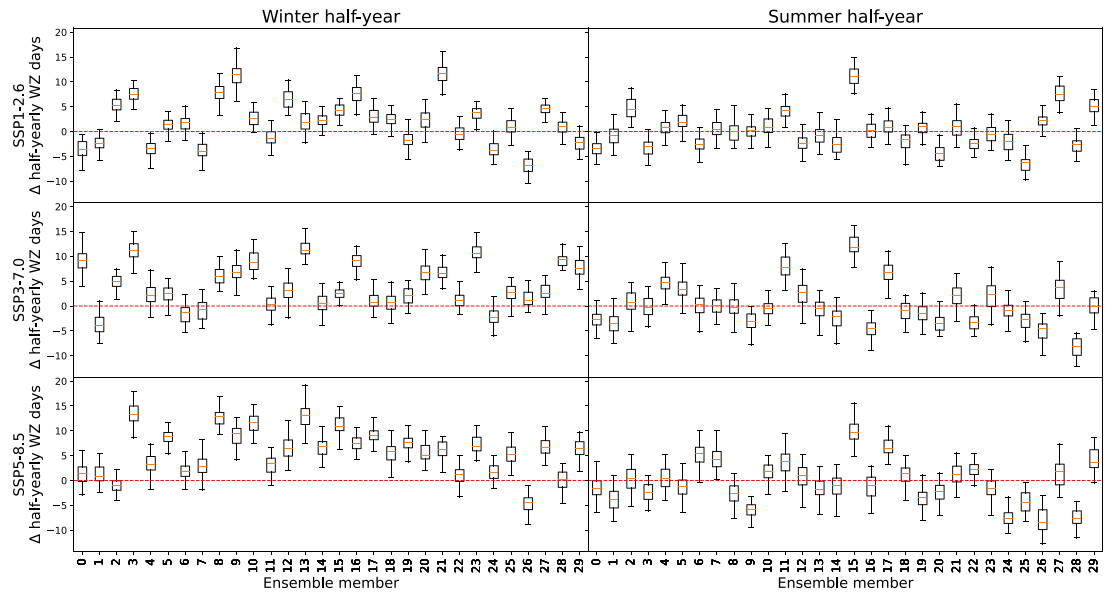


Figure 5. Changes to the number of semi-annual WZ days in MPI-GE for the three future climate change scenarios SSP1-2.6, SSP3-7.0, and SSP5-8.5 for the far future (2071–2100) when compared to their historical reference period (1961–1990). The left-hand side represents the winter half-year (16 October–15 April) while the right side corresponds to the summer half-year (16 April–15 October). Outliers that fall outside the whiskers of the boxplots are excluded from the visualisation.

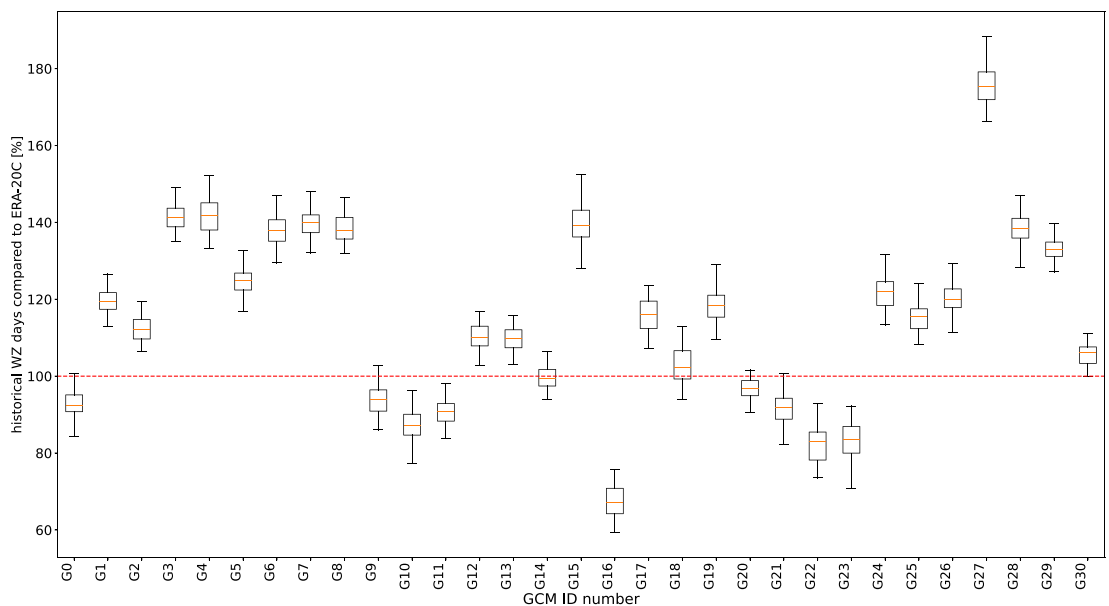


Figure 6. Number of days classified as WZ over the 30 year historical period (1961–1990) for the 31 CMIP6 models (see table 2 for the corresponding names) in percentage of the number classified in ERA-20C for the same time period. The percentages were calculated separately for each of the 50 neural networks. Outliers that fall outside the whiskers of the boxplots are excluded from the visualisation.

period. The standard deviation for the historical reference period over all 1500 data points (30 ensemble members $\times$ 50 networks) for each of the three scenarios is approximately 4.5 d.

4.3. CMIP6 models

A comparison of the number of days classified as WZ in the 31 CMIP6 models with ERA-20C reanalysis data for the historical reference period reveals

significant variation among the models (figure 6). Most models show a higher number of WZ days, with NorESM2-LM (G27) showing the largest divergence of approximately 70%. Many other models overestimate the number of WZ days by around 20%–40%. Nearly a third of the models underestimate the number of WZ days, but usually within a 20% difference for the historical numbers in ERA-20C data. EC-Earth3-Veg-LR (G14), INM-CM5-0 (G17), and

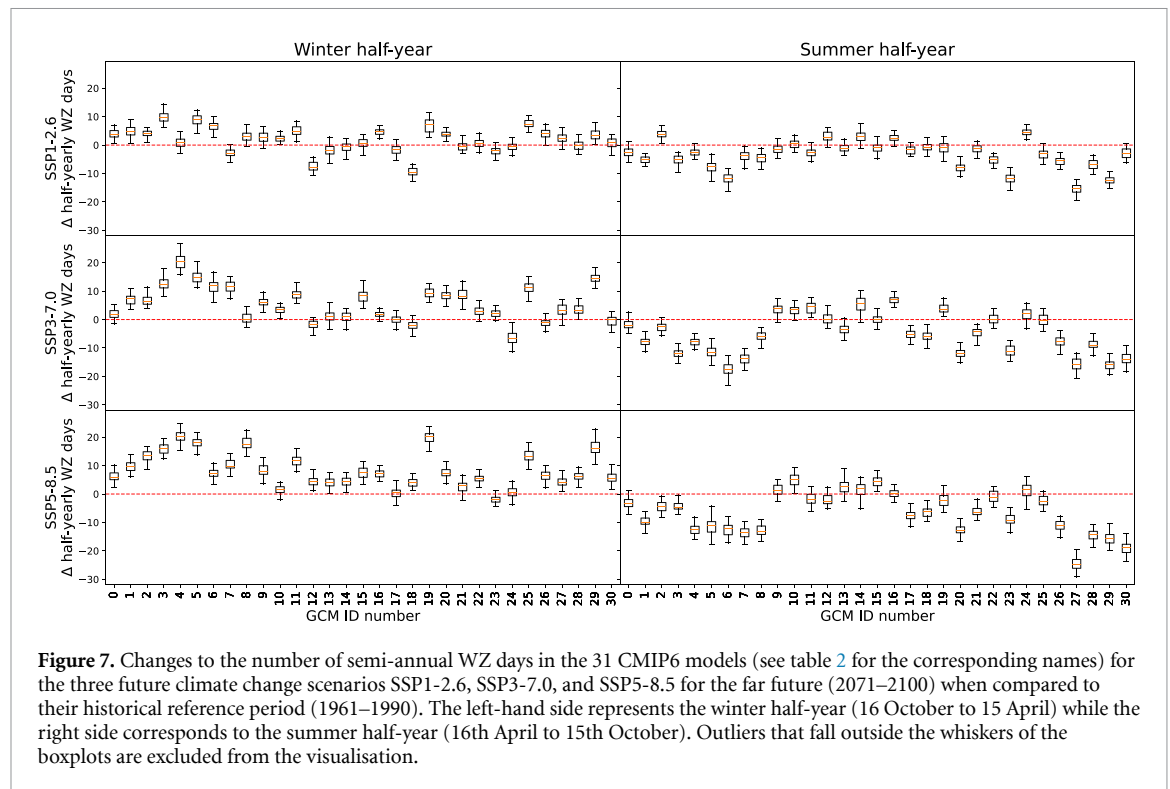


Figure 7. Changes to the number of semi-annual WZ days in the 31 CMIP6 models (see table 2 for the corresponding names) for the three future climate change scenarios SSP1-2.6, SSP3-7.0, and SSP5-8.5 for the far future (2071–2100) when compared to their historical reference period (1961–1990). The left-hand side represents the winter half-year (16 October to 15 April) while the right side corresponds to the summer half-year (16th April to 15th October). Outliers that fall outside the whiskers of the boxplots are excluded from the visualisation.

KACE-1-0-G (G20) are the three models that closely match the historical values.

The 31 CMIP6 models demonstrate clear trends under climate change in the future (2071–2100, figure 7). Like in section 4.2, the year is divided into two halves. The 31 CMIP6 models project an increase in the number of WZ days during the winter half-year. Two-thirds of the CMIP6 members project an increase under scenario SSP1-2.6, with an overall average change of 1.93 additional days of WZ per year during winter. This trend increases under SSP3-7.0 and culminates in SSP5-8.5, where all but one model project an increase. The mean increase across all models is projected to be 8.31 additional WZ days during winter in SSP5-8.5. The majority of models show a steady increase for stronger emission scenarios. However, there are some exceptions, such as ACCESS-CM2, which exhibits a decline in WZ days for SSP3-7.0 before rising again in SSP5-8.5. An increasing number of models project a rise in WZ days in response to a stronger climate change signal, but the numbers projected by the individual models do not necessarily increase significantly. Some models project changes roughly within the same range throughout all three scenarios, while others increase with higher scenarios.

In the summer half-year, the models overall tend to project a decrease in WZ days, but not all point in the same direction. The mean projection for the summer half-year is -3.5 d for SSP1-2.6, and decreasing further in the following scenarios to -5.02 d under SSP3-7.0 and -6.60 d under SSP5-8.5. More than

60% of the models project a decrease for all three scenarios.

5. Discussion

Comparing the performance of our CNMN network architecture with the results of James (2007) and Mittermeier *et al* (2022) showed an improvement in the classification for the majority of weather systems. While the performance is only slightly better for some Großwetterlagen, it was especially enhanced for WZ. The improved classification skills of CNMN can be mostly attributed to the new network architecture, as we aimed to closely replicate the training conditions described in Mittermeier *et al* (2022) to enable a direct comparison. Our CNN is deeper and features 912 K trainable parameters, which is larger than the networks in Mittermeier *et al* (2022), which had 144 K. State-of-the-art networks have significantly improved in classification skill over the last decade, partly due to the increased depth and size of the networks. However, it is important to note that simply increasing the size of the architecture does not lead to better results in this case. We currently lack the necessary training data to prevent overfitting on deeper and wider networks (Liu *et al* 2016). Additionally, many common data augmentation techniques, such as mirroring or rotating the input data, cannot be applied in our study, as the location of, e.g. pressure systems, matters for the accurate classification. Furthermore, adding more input variables did not improve the results, as outlined in section S2.

The analysis of the MPI Grand Ensemble revealed noticeable variability, even among members of the same ensemble. Therefore, it is important to emphasise that the comparison between the frequency of WZ in the ERA20C reanalysis data and the historical CMIP6 runs (figure 6) should not be interpreted as a direct measure of the models' ability to accurately represent weather systems. This is due to the possibility of selecting an ensemble member that performs better or worse than the average ensemble member of a certain model, in cases where multiple runs were available. However, it is evident that some models more closely match the reanalysis data than others. Due to the data augmentation used during the training, it is unlikely that potential biases in the GCM data for mslp and zg500 significantly influence the detection of WZ by CNMN.

Although there are variations in the accuracy of the historical representation, the CMIP6 models and the ensemble members of MPI-GE exhibit clear trends for the winter season. We found that the models project an increase in the number of WZ days during winter. A higher frequency of WZ, associated with strong winds and precipitation, suggests a higher chance for compound flood events in the future. Winter is already the primary season for those events in northern Europe, as this is when storm surges mostly occur (Liu *et al* 2022a). This trend becomes more pronounced under scenarios with stronger climate change. The stronger climate signal leads to a higher signal-to-noise ratio so that the signal of increasing WZ patterns becomes more similar among the different models. Assuming that the internal variability of MPI-GE is similar to the other CMIP6 members, it can be concluded that the overall change signal is greater than the internal variability. These results are based on a wide range of the most recent GCMs, increasing the robustness of this signal.

Both previous studies on future changes to Großwetterlagen found no such changes. James (2006), using their composite method, attributed this to the internal variability of the HadGEM1 model, which they had used for their analysis. However, we have overcome this limitation with 31 GCMs. Another important factor are the significant advances that GCMs have made since their work was published nearly two decades ago. Mittermeier *et al* (2022) reported no median change in the number of WZ days during winter under SSP3-7.0 for a SMILE, whereas our work assessed a larger variety of models.

Other studies utilised different classification methods to investigate changes to atmospheric circulation. Some studies, like Huguenin *et al* (2020) and Hansen *et al* (2023), found no increase in westerly flow circulation types. Conversely, several studies have reported an increase in westerly circulation types, which is consistent with the findings of the present study. For instance, Hoy *et al* (2014) found that westerly inflow has increased in the recent past,

leading to higher precipitation in northern Europe, and Rohrer *et al* (2017) reporting an increase in westerlies until the end of the current century for the Alpine region. Similarly, Cahynová and Huth (2016) categorised mslp fields over Europe into 24 circulation types and reported an increasing frequency of westerly types. Demuzere *et al* (2009) also reported similar changes in Europe under climate change, using the Lamb weather types for the ECHAM5-MPI/OM climate model.

For the summer season, a contrary trend has been noted, with the majority of models projecting a decrease in WZ days. While the overall mean of the CMIP6 models shifts, some models project little to no change, or even an increase under SSP5-8.5. This is in contrast to the winter season where almost all GCMs project a shift in the same direction. These results are consistent with the findings of Otero *et al* (2018), who analysed ten CMIP5 GCMs and discovered a decrease in westerlies during the summer months under SSP8-5.8. Similarly, Herrera-Lormendez *et al* (2023) found the same trend in the analysis of 21 CMIP6 models.

The reasons behind changes in circulation patterns due to climate change remain a topic of ongoing scientific investigation (Perlwitz *et al* 2017), with uncertainty arising from low-frequency chaotic variability and the sensitivity of circulation patterns to climate forcing in the models (Shepherd 2014). However, our study indicates a significant increase in the frequency of WZs during the winter season. Investigating the exact reasons for this is beyond the scope of the present study and should be analysed in more detail in a future study.

The results of this study naturally come with some caveats. Although our network's accuracy and F1-score have improved compared to previous work, they are not flawless. Moreover, the CNN's overall accuracy is inherently constrained by the subjective human influence on the classification in the Großwetterlagen catalogue and the 3 d rule. If, for any reason, a bias was present in the classification, our CNNs would inevitably learn it, as they aim to mimic the human classification, which serves as the 'ground truth'. When comparing the historical reference period of the GCMs to the ERA-20C reanalysis data, it became apparent that some models have difficulties with correctly representing the frequency of WZ. Furthermore, there is a certain amount of variation even among ensemble members, as shown for MPI-GE. Unfortunately, many GCMs provide only one or a limited number of ensemble members with the required three scenarios and variables, making it impossible to assess the variation among them.

There are several interesting directions that future research can take to improve on the results presented in this study. One possibility to improve the classification is the construction of deeper architectures, that are at the same time less prone to overfitting. In addition to data augmentation, there are

other techniques such as gradient penalties, semi-supervised pre-training, channel shuffle, and many others that are used to reduce overfitting during training, but were beyond the scope of this study. Another option is to consider the temporal aspect of the data which has so far not been utilised. Currently, classification is based on the mean of a single day, but with ERA-20C's 3 h data, it could be classified like a video, which is simply a sequence of images. While there are networks available for classifying video clips, they may be too large and prone to overfitting. It is important to note that, for a study similar to ours, collecting the necessary CMIP6 data on a sub-daily scale may be difficult as they are not readily available for most GCMs.

6. Conclusions

In this study, we utilised a convolutional architecture that improved the automatic classification of Großwetterlagen over Europe. This trained network was then utilised to analyse future changes under three climate change scenarios in 31 CMIP6 models and the MPI-GE. This is one of the first studies to conduct weather pattern classification for Europe for a large CMIP6 ensemble. Our analysis showed that there is a certain internal variability among the ensemble members of MPI-GE caused by the different initialisations, with some members deviating from the general trend of more WZ days in winter. Although the CMIP6 models are the latest generation of GCMs, only a few match closely the frequency of WZ for the historical reanalysis data of ERA-20C. Nevertheless, the models clearly showed a rise in WZ days during the winter season, but a decline in the summer.

Projected changes in WZ days are in line with the general consensus of hotter and dryer summers, while the winter will become wetter. The strength of the change increases under stronger climate change scenarios; from almost no signal under SSP1-2.6 to a very strong signal under RCP5-8.5. As winter is already the primary season for compound flood events, an increase in WZ in the future suggests a potentially higher number of such events. These changes, combined with sea level rise, will pose significant challenges for coastal protection in the future.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://doi.org/10.5281/zenodo.14608761>.

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Philipp Heinrich developed the analysis methods, performed the data analysis and wrote the manuscript. Ralf Weisse and Stefan Hagemann supervised

the research activities, discussed the outline of the study and its results, revised the manuscript, and contributed to the interpretation of the results.

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Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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